

Design of a Flexible, Zn-Selective Protein Scaffold that Displays Anti-Irving–Williams Behavior

Tae Su Choi and F. Akif Tezcan*



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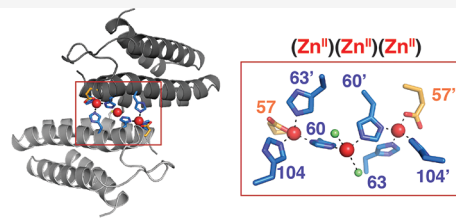


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ABSTRACT: Selective metal binding is a key requirement not only for the functions of natural metalloproteins but also for the potential applications of artificial metalloproteins in heterogeneous environments such as cells and environmental samples. The selection of transition-metal ions through protein design can, in principle, be achieved through the appropriate choice and the precise positioning of amino acids that comprise the primary metal coordination sphere. However, this task is made difficult by the intrinsic flexibility of proteins and the fact that protein design approaches generally lack the sub-Å precision required for the steric selection of metal ions. We recently introduced a flexible/probabilistic protein design strategy (MASCoT) that allows metal ions to search for optimal coordination geometry within a flexible, yet covalently constrained dimer interface. In an earlier proof-of-principle study, we used MASCoT to generate an artificial metalloprotein dimer, (AB)₂, which selectively bound Co^{II} and Ni^{II} over Cu^{II} (as well as other first-row transition-metal ions) through the imposition of a rigid octahedral coordination geometry, thus countering the Irving–Williams trend. In this study, we set out to redesign (AB)₂ to examine the applicability of MASCoT to the selective binding of other metal ions. We report here the design and characterization of a new flexible protein dimer, B₂, which displays Zn^{II} selectivity over all other tested metal ions including Cu^{II} both *in vitro* and *in cellulo*. Selective, anti-Irving–Williams Zn^{II} binding by B₂ is achieved through the formation of a unique trinuclear Zn coordination motif in which His and Glu residues are rigidly placed in a tetrahedral geometry. These results highlight the utility of protein flexibility in the design and discovery of selective binding motifs.



Metal Selectivity through Flexible Protein Design

INTRODUCTION

Up to a third of all proteins are associated with metal ions,^{1–3} which are often necessary for the proper folding/assembly of proteins^{4–7} and enable them to fulfill essential functions such as signaling,^{8–10} small molecule transport,^{11–13} and catalysis.^{14–18} The correct pairing of metalloproteins with their cognate metal ions is essential for these cellular functions.¹⁹ In fact, there are very few known cases where the natural metal component of a protein can be substituted by another metal ion *in vivo* without deleterious consequences.^{20–25} Despite the fact that there is a limited set of metal-coordinating amino acid residues (e.g., His, Asp, Glu, Cys, Met, Tyr), three-dimensional protein structures can achieve some extent of metal selectivity through a suitable combination of such residues in the primary coordination sphere in addition to the further tuning of metal–ligand interactions through secondary sphere effects.^{19,26–28} Of particular significance is steric selection by proteins, which describes the precise spatial arrangement of the primary-sphere residues to discriminate between metal ions (particularly, those of the d-block) based on their ionic radii and their electronic preferences for specific coordination geometries.¹⁰ Yet, all proteins are inherently flexible, meaning that metal binding can perturb the intended coordination geometries of the primary-sphere residues such that steric selection is often overridden.^{29–34} As a result, metal binding affinities of proteins are largely governed by the electronic properties of the metal

ions themselves and almost universally follow the Irving–Williams (IW) series (Mn^{II} < Fe^{II} < Co^{II} < Ni^{II} < Cu^{II} > Zn^{II})^{29,35–38} with rare exceptions.^{28,39} For instance, most metalloproteins coordinate Cu^{II}—the most dominant member of the IW series—with binding constants that are typically orders of magnitude higher than those of their cognate metal ions.^{26,37,40–45} Thus, as a necessity to overcome the restrictions set forth by the IW series, living organisms have evolved sophisticated systems for metal homeostasis and speciation,^{46,47} which actively control concentrations of transition-metal ions in cells, use specialized proteins for metal transport and delivery,^{48–51} and compartmentalize folding and metalation.²⁶ In fact, the availability of exchangeable metal ions in cells is tightly regulated by an array of specialized metallosensor proteins and gene transcription factors^{52–55} such that the metal concentrations follow the inverse of the IW series (i.e., Mn^{II} > Fe^{II} > Co^{II} > Ni^{II} > Cu^{II} < Zn^{II}) to avoid the

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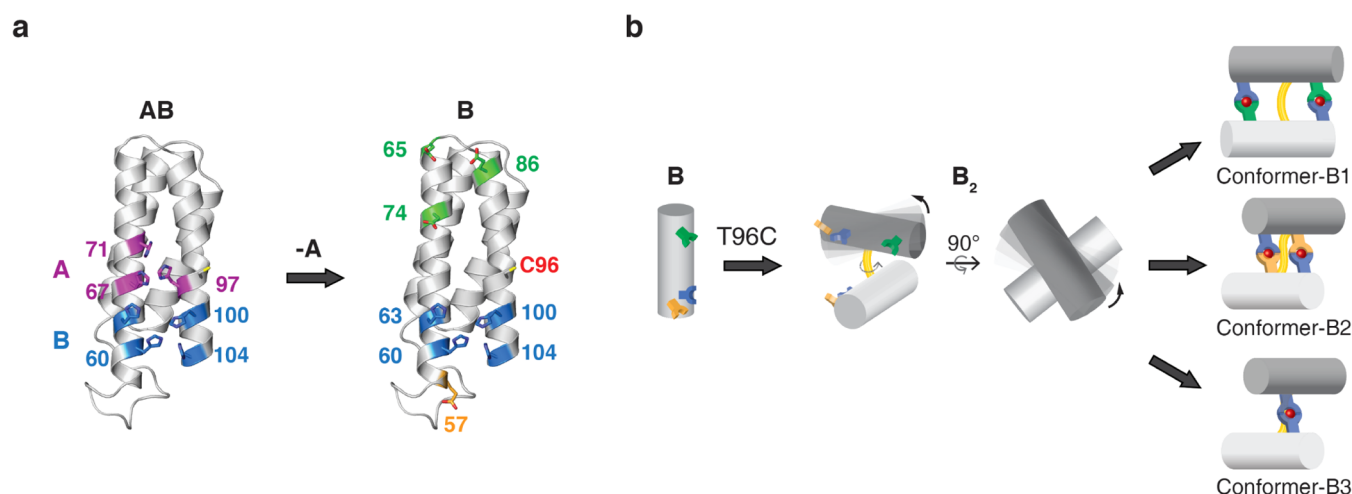


Figure 1. (a) Position of metal-binding residues in the B construct remodeled from the His-rich AB variant. Purple or blue sticks represent His residues in Site A (H67/H71/H97) or in Site B (H60/H100/H104) and native H63. Yellow (E57) and green (D65/D74/E86) sticks indicate Asp or Glu residues in wild-type cytochrome *cb₅₆₂*. (b) Metal-induced conformational changes of a disulfide-bond-linked B_2 interface. Locations of potential metal-binding residues can induce variations in the metal-bound conformations.

nondiscriminate binding of the IW-dominant metal ions to cellular proteins.³⁶

While nature has—by necessity—resorted to such active control mechanisms to ensure correct metalation of proteins in cells, it should still be possible to thermodynamically overcome the IW restrictions by the proper design of protein scaffolds that can rigidly impose steric selection.⁵⁶ Potentially, such artificial metalloproteins could not only be incorporated into cells to perform *in vivo* tasks (e.g., catalysis,⁵⁷ control and perturbation of metal homeostasis^{56,58}) but they could also serve as metal sequestration and remediation agents in environmental/*ex vivo* settings.⁵⁹ However, the *de novo* design of a selective transition-metal-binding protein remains difficult because no current computational design approach possesses the structural precision (on the order of tenths of an Ångström) required to sterically distinguish between transition-metal ions.^{60,61} To address this challenge, we recently introduced a probabilistic design strategy termed Metal Active Sites through Covalent Tethering (MASCOT).⁴¹ In MASCOT, two monomers of a protein are each surface-engineered with a Cys residue to covalently link them by a disulfide bond. Combinations of metal-coordinating residues flanking the Cys residue are then added to furnish a flexible protein homodimer capable of stably coordinating metal ions in the disulfide-linked interface. Such a design offers several advantages: (1) the disulfide bond is sufficiently flexible to allow the protein dimer to explore a wide range of conformational states (*i.e.*, the dimer possesses a shallow energy landscape) in the “search” for a thermodynamically favored metal coordination mode, but also short enough such that the resulting metal-bound dimer is structurally constrained (*i.e.*, it represents a distinct free-energy minimum) and the location of metal binding is largely limited to the disulfide-linked interface; (2) the inherent symmetry of the system can enable the binding of multiple metal ions in the interface, thus giving rise to cooperative binding and mutual rigidification; (3) the surface-exposed nature of the interfacial metal coordination sites ensures that bound metal ions can exchange freely with unbound species so that kinetically trapped species are avoided (*i.e.*, the system is under thermodynamic control).

Using the four-helix bundle protein cytochrome *cb₅₆₂* as a building block for MASCoT, we recently engineered an artificial homodimeric scaffold termed $(AB)_2$ with multiple His-rich motifs surrounding the interfacial disulfide linkage (Figure 1a).⁵⁶ We found that the flexible $(AB)_2$ scaffold could adopt at least two mutually exclusive conformational states depending on the identity of the bound metal ion. Detailed characterization of these conformations led to the discovery of two cooperative metal coordination motifs that consisted of five histidines and one aqua ligand ($5\text{His}/1\text{H}_2\text{O}$) in a rigidified octahedral geometry, which selectively bound Co^{II} or Ni^{II} and sterically disfavored Cu^{II} coordination, thus thermodynamically overcoming the IW series. Following up on that proof-of-principle study, we asked whether our strategy could be applied to discover other selective metal-binding motifs. Accordingly, we reengineered the flexible $(AB)_2$ dimer interface with alternative coordination sites.

We targeted Zn^{II} selectivity over other metal ions in the IW series (including Cu^{II}) and heavy-metal ions, given the diverse structural and functional roles Zn^{II} ions play in natural metalloproteins,^{27,62–66} and the vulnerability of natural Zn^{II} coordination sites to deactivation by heavier metal ions (e.g., Cd^{II} , Pb^{II}).^{67–69} Indeed, the design of selective Zn^{II} chelators has been broadly pursued using a variety of molecular and macromolecular platforms such as fluorogenic ligands,^{70,71} peptoids,⁷² and proteins.^{58,73} We report here the probabilistic design and the structural, biophysical, and biochemical characterization of a reengineered dimeric assembly, B_2 , which displays Zn^{II} selectivity over all other tested metal ions including Cu^{II} . Such selective, anti-IW Zn^{II} binding is achieved thanks to the formation of a unique trinuclear Zn coordination motif consisting of His and Glu residues arranged in a tetrahedral geometry. Our results further establish structural flexibility as an invaluable protein design parameter, particularly with respect to the discovery of metal coordination motifs that would be difficult to build from first principles.

RESULTS AND DISCUSSION

Design and Characterization of B_2 . The previously designed $(AB)_2$ dimer is built from two identical monomers of

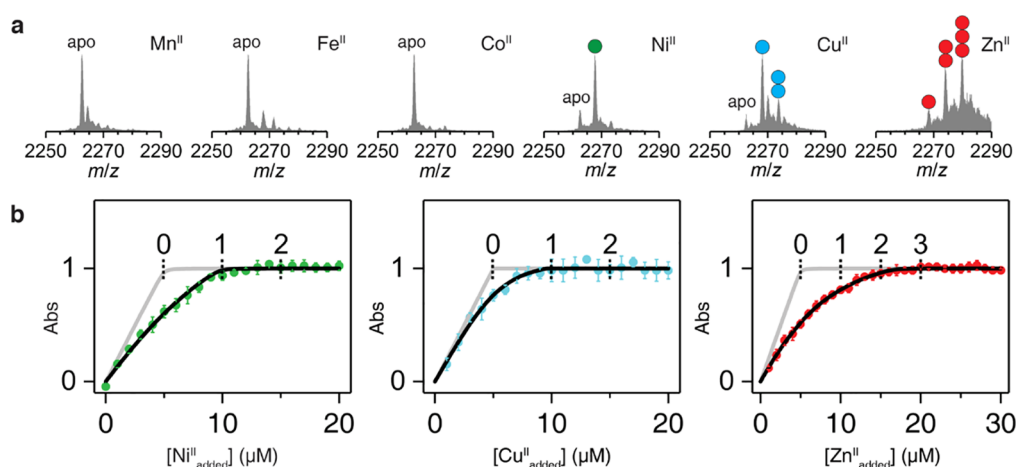


Figure 2. (a) ESI-MS spectra of B_2 in the presence of Mn^{II} , Fe^{II} , Co^{II} , Ni^{II} , Cu^{II} , and Zn^{II} . The color of the circles represents Ni^{II} (green), Cu^{II} (cyan), or Zn^{II} (red) ion bound to B_2 . The number of circles indicates the number of metal ions bound to B_2 . (b) Competitive metal-binding titrations of Ni^{II} (left), Cu^{II} (middle), and Zn^{II} (right) using Fura-2. Changes in normalized Fura-2 absorbance at 335 nm (green, cyan, and red data points and black fits) are plotted along with theoretical metal-binding isotherms (gray) in the absence of B_2 . Simulated equivalence points (0–3 equiv) for metal binding are indicated on the theoretical metal-binding isotherms. Experimental data points and error bars are presented as mean and standard deviation of three independent measurements.

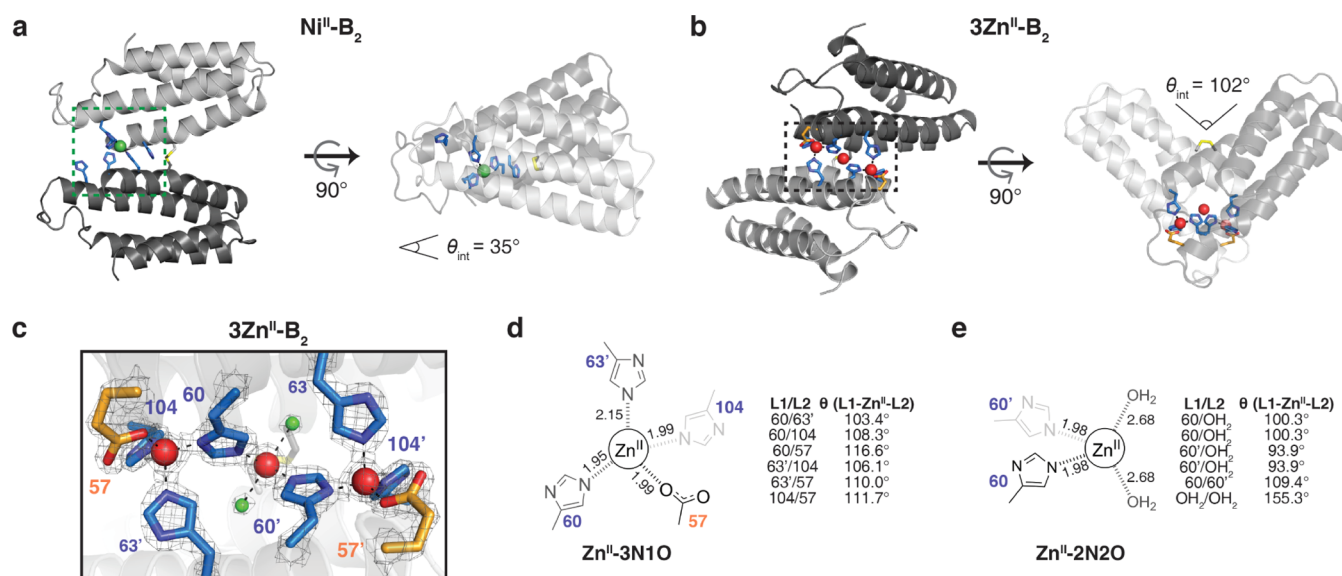


Figure 3. Crystal structures of (a) $1Ni^{II}-B_2$ (PDB: 8DRD) and (b) $3Zn^{II}-B_2$ (PDB: 8DRF). Green and red spheres represent Ni^{II} and Zn^{II} ions, respectively. Close-up view of the green boxed area is presented in Figure S2d. (c) Close-up of the primary coordination spheres of trinuclear Zn^{II} -binding sites. Green spheres represent water molecules. Contour levels of $2mF_o-DF_c$ (gray mesh) electron density are 5.0σ for Zn^{II} and 2.0σ for C, N, and O atoms. (d, e) Coordination distances and angles between Zn^{II} and ligands: (d) peripheral Zn^{II} sites coordinated to E57/H60/H104/H63' and (e) central Zn^{II} site coordinated to H60/H60'/ $2H_2O$.

cytochrome cb_{562} containing two 3His motifs in two surface sites: Site A (H67/H71/H97) is immediately next to the Cys96 (for covalent dimerization) and Site B (H60/H100/H104) is located two-to-three helix turns away (Figure 1a).⁵⁶ As in the cases of Co^{II} and Ni^{II} , competitive titrations indicated that $(AB)_2$ also bound approximately 2 equiv of Zn^{II} with low nM dissociation constants (Figure S1). Yet, unlike Co^{II} or Ni^{II} , Zn^{II} could not outcompete Cu^{II} : native electrospray ionization mass spectrometry (ESI-MS) experiments indicated that $(AB)_2$ preferred to form a $1Cu^{II}$ or a $1Zn^{II}:1Cu^{II}$ heterometallic complex when both metal ions were present in solution.⁵⁶ We attributed the noncompetitiveness of Zn^{II} to the fact that it is almost never found in 5- or 6-coordinate, all-His coordination geometries in proteins.⁶³ Thus, the $2Zn^{II}$ -bound form of $(AB)_2$

likely does not adopt the same conformation as the $2Co^{II}$ - and $2Ni^{II}$ -bound forms. This observation motivated us to change the location and type of metal-binding residues to induce the formation of new Zn^{II} -binding sites in the dimer interface. Hence, we eliminated the 3His motif in Site A in the AB construct, converting it to the native residues (I67, Q71, T97) of cytochrome cb_{562} (Figure 1a). We reasoned that in the resulting construct ("B"), the 3His motif of Site B (in combination with the native H63) could pair with one of the native Glu and Asp residues (E57, D65, D74, or E86) from the opposing monomer to form stable, tetrahedral Zn^{II} coordination motifs (Figure 1a). In light of the expected flexibility of the B_2 dimer and the placement of metal-coordinating residues, we predicted that at least two different conformations could

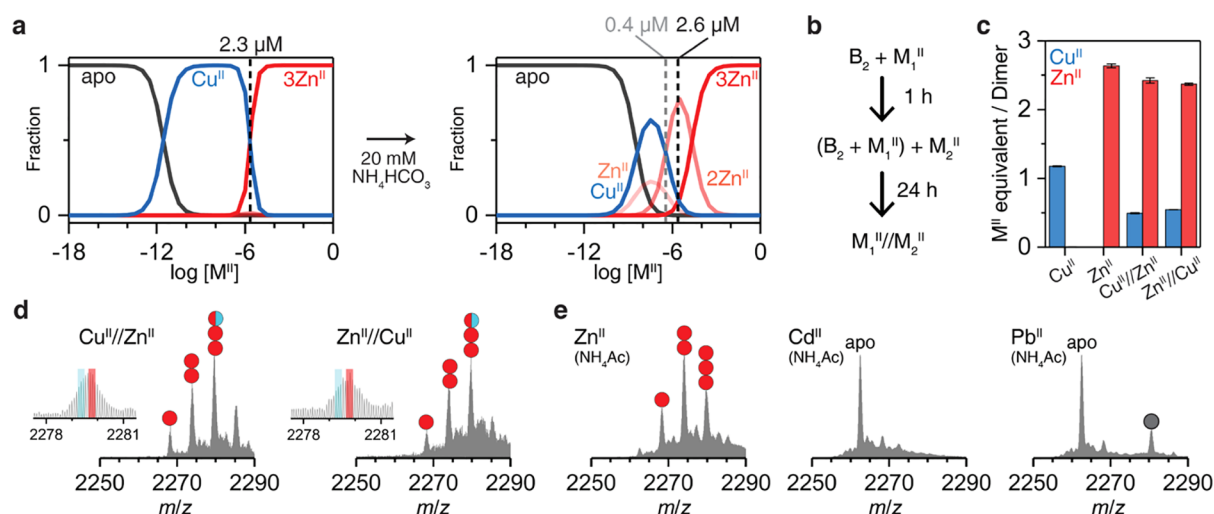


Figure 4. (a) Theoretical fractionation profiles of B_2 , Cu^{II} - B_2 , and $3Zn^{II}$ - B_2 species as functions of buffered Cu^{II} and Zn^{II} concentrations in 20 mM MOPS and 150 mM NaCl (left) and 20 mM NH_4HCO_3 (right). The dotted line represents the concentration at which the portion of $3Zn^{II}$ - B_2 or $2Zn^{II}$ - B_2 is equal to Cu^{II} - B_2 . (b) Sample preparation scheme for metal-binding competition. If Cu^{II} is first added to B_2 and Zn^{II} is added later, the sample is labeled Cu^{II}/Zn^{II} . (c) ICP-MS analysis of metal contents in B_2 solutions. Using Fe ions of the heme group as an internal standard, the M^{II} equivalent per B_2 was evaluated. Under both Cu^{II}/Zn^{II} and Zn^{II}/Cu^{II} conditions, Zn^{II} outcompetes Cu^{II} . Experimental data points and error bars are presented as mean and standard deviation of three independent measurements. (d) ESI-MS spectra of B_2 in Cu^{II}/Zn^{II} (left) and Zn^{II}/Cu^{II} (right). Filled circles represent the equivalents of Cu^{II} (cyan) and Zn^{II} (red) ions bound to B_2 . Half-circles indicate a mixture of Cu^{II} and Zn^{II} complexes. Spectra in insets correspond to expanded m/z ranges of the $3M^{II}$ - B_2 species. (e) ESI-MS spectra of B_2 obtained in 20 mM NH_4Ac (pH 6.8). Circles in ESI-MS spectra represent the number of Zn^{II} (red) and Pb^{II} (dark gray) ions bound to B_2 .

ensue upon binding two metal ions: the antiparallel Conformer-1 or V-shaped Conformer-2 (Figure 1b). In addition, we expected that a third, single-metal-bound structure (Conformer-3) might also form, using only the His residues of the B sites.

We first characterized the structure and metal-binding properties of B_2 in solution. Analytical ultracentrifugation (AUC) measurements indicated that B_2 is a stable dimer both in the absence and presence of metal ions (Figure S2a). We performed ESI-MS experiments on mixtures of B_2 (5 μM) and up to 3 equiv of mid-to-late first-row transition-metal ions (Mn^{II} , Fe^{II} , Co^{II} , Ni^{II} , Zn^{II} , Cu^{II}) to determine whether B_2 forms stable metal complexes in solution (Figure 2a). In the absence of metal ions, B_2 showed a broad charge state distribution, from +9 to +24 (Figure S2b). We chose the most abundant +11 charge state as the representative peak for metal-binding analysis.⁵⁶ The mass spectra for the Mn^{II} , Fe^{II} , and Co^{II} samples were dominated by the metal-free (apo) form of B_2 (Figure 2a), consistent with the low expected affinities of these metal ions according to the IW series. In contrast, we observed dominant peaks for $1Ni^{II}$ -, $1Cu^{II}$ -, and $3Zn^{II}$ -bound B_2 . Competition titrations using Fura-2 as a chromogenic indicator confirmed that B_2 tightly bound one Ni^{II} ($K_d = 15$ nM), three Zn^{II} ($K_{d,aver.} = 34$ nM) and one Cu^{II} ion ($K_d = 3$ pM) (Figure 2b and Table S1). Notably, both the Ni^{II} and Cu^{II} affinities of B_2 were considerably lower (3- to 10-fold) than those of $(AB)_2$ (~ 5 nM for Ni^{II} , 0.2 pM for Cu^{II}),^{41,56} implying that the removal of the Site A motif likely resulted in low-coordinate (*i.e.*, $n < 5$ or 6) Ni- and Cu-binding modes in the dimer interface.

Next, we set out to crystallize B_2 in the presence of first-row transition-metal ions and determine the structures of metal-bound forms of B_2 . Mn^{II} , Fe^{II} , and Co^{II} samples yielded no crystals, whereas we were able to obtain diffraction-quality crystals with Ni^{II} , Cu^{II} , and Zn^{II} . Interestingly, the diffraction analysis of the Cu^{II} - B_2 cocrystals revealed an apo- B_2 structure

(2.4 Å resolution) (Figure S2c) despite the pM dissociation constant of the corresponding complex. Possible explanations include the presence of multiple Cu^{II} -bound B_2 conformations none of which are thermodynamically dominant or crystal packing effects that render the Cu^{II} - B_2 complex unfavorable compared to apo- B_2 .

By contrast, the structures of both Ni^{II} - and Zn^{II} adducts of B_2 accorded with the solution studies. The $1Ni^{II}$ -bound B_2 crystal structure (1.89 Å resolution) featured a parallel and slightly tilted arrangement (interfacial angle $\theta_{int} = 35^\circ$; Figure 3a) of the disulfide-linked protein monomers, approximating one of the expected dimer conformations, namely, Conformer-3 (Figure 1b). The Ni^{II} center is coordinated by four His residues (H60/H100 from one monomer and H100'/H104' from the other) and two water molecules (Figure S2d), in agreement with its lower stability compared to the 5His coordination sites in $(AB)_2$. The structure of the $3Zn^{II}$ - B_2 complex (1.7 Å resolution) also revealed a parallel dimer arrangement, but quite distorted from that of the Ni^{II} - B_2 assembly and adopting a V-shaped architecture ($\theta_{int} = 102^\circ$; Figure 3b), consistent with Conformer-2 (Figure 1b). This conformation is induced by an unusual coordination geometry of three Zn^{II} ions. Two of the Zn^{II} ions are equivalent and related to one another by the C_2 symmetry of the protein dimer (Figure 3c). They are coordinated to three His residues (H60/H63/H104) from one monomer and a Glu (E57) from the second, yielding a nearly ideal tetrahedral geometry (avg. bond angle = $109.4 \pm 4.6^\circ$, avg. bond distance = 2.0 ± 0.1 Å, Figure 3d). These coordination sites are directly bridged by the third Zn^{II} ion, which coordinates the N_ϵ atoms of H60 residues along with two water molecules (also in a tetrahedral geometry, Figure 3e), thus completing a cooperative and highly rigidified metal coordination network (Figure S3).

Solution Analysis of the Zn^{II} Selectivity of B_2 . Having determined the Zn^{II} binding properties of B_2 , we examined if it also displays Zn^{II} selectivity over Cu^{II} , which would represent

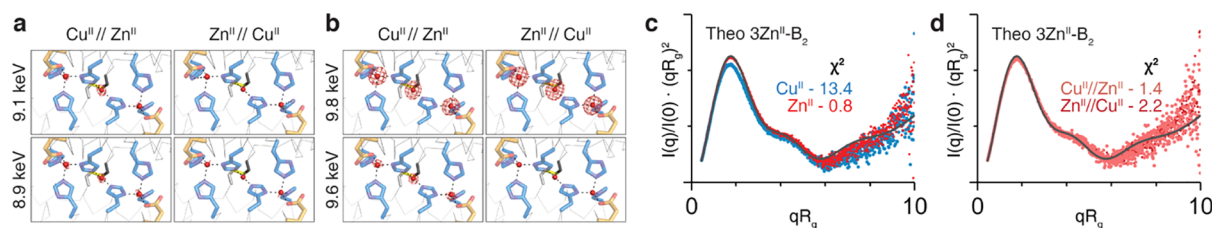


Figure 5. (a, b) Anomalous scattering densities of metal ions in the crystal structures of B₂ formed in Cu^{II}//Zn^{II} or Zn^{II}//Cu^{II} conditions. Anomalous densities contoured at 5.0 σ were collected around the K-edge of (a) Cu (~9.0 keV) or (b) Zn (~9.7 keV). Changes in anomalous densities above (top) and below (bottom) the K-edge of Zn^{II} indicate that the metal-binding sites in the B₂ interface are occupied by Zn^{II} ions. (c) Normalized Kratky plots of B₂ in the presence of Cu^{II} (cyan) or Zn^{II} (red) compared with the theoretical plot of 3Zn^{II}-B₂ (black) at 200 μ M dimer concentration ($[M^{II}] = 200 \mu\text{M}$). (d) Normalized Kratky plot of B₂ in Cu^{II}//Zn^{II} and Zn^{II}//Cu^{II} conditions presented with the theoretical plot of 3Zn^{II}-B₂ (black) at 200 μ M dimer concentration ($[M_1^{II}]/[M_2^{II}] = 200 // 200 \mu\text{M}$). All log-scale SAXS plots are shown in Figure S6.

anti-IW behavior. Although the per-site binding affinity of the Cu^{II} for B₂ is >1,000-fold higher than that for the individual Zn^{II} sites, we hypothesized that the higher valency and cooperativity of the trinuclear Zn^{II} sites would enable the out-competition of Cu^{II}. Indeed, when the experimentally determined equilibrium constants were used to calculate theoretical fractions of B₂ and M^{II}-B₂ species ($M^{II} = \text{Cu}^{II}$ or Zn^{II}), the 3Zn^{II}-B₂ complex was predicted to dominate 1Cu^{II}-B₂ species at $[M^{II}] \geq 2.3 \mu\text{M}$ (Figure 4a, see the Supporting Methods for calculations).

We first employed inductively coupled plasma mass spectrometry (ICP-MS) to assess competitive metal binding to B₂. For these experiments, mixtures of B₂ (5 μM dimer) with various metal compositions were prepared in 20 mM MOPS (pH 7.4) solutions: (1) “noncompetitive” samples containing 15 μM Cu^{II} or 15 μM Zn^{II} only; (2) “competitive” samples containing 15 μM Cu^{II} + 15 μM Zn^{II}. The latter samples were prepared in two different ways to ensure that competition occurred under thermodynamic control, in that the protein solution was first incubated either with the competitive metal or Zn^{II}, followed by the addition of Zn^{II} or the competitive metal (Figure 4b). Subsequently, unbound metal ions in the protein solutions were washed out through a buffer exchange prior to ICP-MS measurements. Under noncompetitive conditions, the B₂ dimer associated with 1.1 equiv Cu^{II} or 2.6 equiv Zn^{II} (Figure 4c), which are similar to the expected stoichiometries. Under both competitive conditions (Cu^{II}//Zn^{II} or Zn^{II}//Cu^{II}), we observed that ~2.5 equiv. Zn^{II} and ~0.5 equiv Cu^{II} bound the B₂ dimer, consistent with the dominance of Zn^{II} binding over Cu^{II}. We note that Cu^{II} is frequently bound nonspecifically to protein surfaces,^{41,56,73} which may explain its substoichiometric association with B₂ under competitive conditions.

We next turned to ESI-MS to investigate distributions of metalated B₂ species under competitive conditions (Figure 4d). We used the same solution conditions as above, except that we included the volatile buffering agent ammonium bicarbonate NH₄HCO₃ (20 mM, pH 7.8) which is compatible with ESI experiments (instead of 20 mM MOPS). Because NH₄HCO₃ is capable of metal coordination, we had to account for the fact that the apparent K_d values of the Cu^{II} and Zn^{II}-B₂ complexes would be increased (*i.e.*, weaker binding) due to the formation of metal-ammine complexes.⁷⁴ Indeed, competitive Fura-2 titrations in 20 mM NH₄HCO₃ solutions indicated up to ~300-fold weaker B₂ binding for Cu^{II} and Zn^{II} (particularly for the third coordination site) compared to the affinities determined in the presence of MOPS (Figure S4a,b). The revised metal fractionation profiles indicated that 2Zn^{II}-B₂

species would be the dominant species over the 1Cu^{II}-B₂ complex starting at $[M^{II}] \approx 0.4 \mu\text{M}$, followed by the increasing population of the 3Zn^{II}-B₂ complex at $[M^{II}] \approx 2.6 \mu\text{M}$ (Figure 4a). Indeed, the ESI-MS spectra revealed that 3Zn^{II}-B₂, 2Zn^{II}-B₂, and 2Zn^{II}+1Cu^{II}-B₂ species were predominant under both Cu^{II}//Zn^{II} and Zn^{II}//Cu^{II} conditions (Figure 4d), with no peak detected for the 1Cu^{II}-B₂ complex. As a control, we also conducted ESI-MS experiments to assess Zn^{II} vs. Ni^{II} competition for B₂ binding, with the expectation that the latter should not be able to compete with the former even at sub- μM levels based on the experimentally determined binding affinities. As expected, both Ni^{II}//Zn^{II} and Zn^{II}//Ni^{II} samples only displayed ESI-MS peaks for the 3Zn^{II}-B₂ complex (Figure S4c).

Finally, in addition to Cu^{II} and the metal ions in the IW series, Cd^{II} and Pb^{II} with closed-shell electronic configurations like Zn^{II} have been reported as competitors to Zn^{II} coordination sites in metalloproteins.^{67–69} To test Zn^{II} selectivity over Cd^{II} and Pb^{II}, the binding between B₂ (5 μM) and Cd^{II} (15 μM) or Pb^{II} (15 μM) was examined using ESI-MS in 20 mM ammonium acetate (NH₄Ac) (pH 6.8). This buffer agent was chosen instead of NH₄HCO₃ because Cd^{II} and Pb^{II} form insoluble salts with carbonate (CO₃²⁻).⁷⁵ Cd^{II} and Pb^{II} did not form a stable complex with B₂, while Zn^{II} binding was maintained (Figure 4e). Moreover, in the mixtures of Zn^{II} and Cd^{II} or Pb^{II}, Zn^{II} always outcompeted other metal ions (Figure S4d,e). Compared to Zn^{II} (bond distance = $2.1 \pm 0.1 \text{ \AA}$ for N donors, $2.3 \pm 0.3 \text{ \AA}$ for O donors), both Cd^{II} and Pb^{II} form coordination complexes with considerably longer bond distances ($2.4 \pm 0.3 \text{ \AA}$ for N donor, $2.6 \pm 0.2 \text{ \AA}$ for O donor).^{76,77} We surmise that B₂ cannot adopt the Conformer-2 dimer geometry while also accommodating such long coordination bond distances within the observed trinuclear coordination motif. Taken together, our analytical measurements show that B₂ is selective for Zn^{II} coordination over all relevant metal ions including Cu^{II}.

Structural Analysis of the Zn^{II} Selectivity of B₂. To examine Zn^{II} vs. Cu^{II} competition for B₂ binding at higher metal/protein concentrations and to obtain direct structural information, we carried out competitive co-crystallization studies. In these studies, a solution of 4 mM B₂ was first incubated with 6 mM Cu^{II} or Zn^{II} for 1 h, followed by incubation with the second metal ion (Zn^{II} or Cu^{II}) for 24 h to reach thermodynamic equilibrium (Figure 2b). These stock solutions were used for sitting-drop crystallization and the harvested crystals were used for X-ray diffraction analysis. Under both conditions (Cu^{II}//Zn^{II} and Zn^{II}//Cu^{II}), the observed crystal structures (1.65 and 1.55 $\text{\AA}$ resolution,

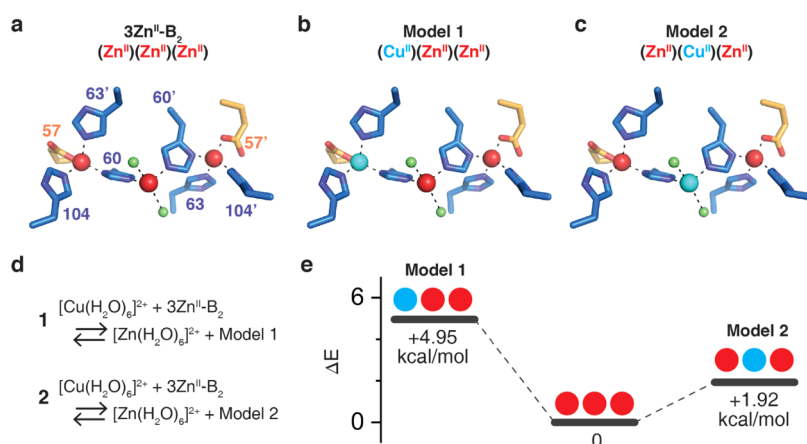


Figure 6. (a–c) Metal configurations of trinuclear metal-binding sites used for 6-31g+(d,p) DFT calculations. Positions of metal ions and metal-binding ligands were extracted from the crystal structure and fixed during DFT calculations. Red, cyan, and green spheres represent Zn^{II} , Cu^{II} , and water, respectively. In Models 1 and 2 (b, c), one of the Zn^{II} ions in (a) was replaced with Cu^{II} based on the XYZ positions of Zn^{II} . (d) Metal exchange reactions between the 3Zn^{II} configuration and model structures. To calculate energy changes during metal exchanges, free Cu^{II} and Zn^{II} ions were considered as aquo complexes. (e) Metal-dependent energy changes between Models 1 and 2 (b, c) and $3\text{Zn}^{\text{II}}\text{-B}_2$ (a). On the y-axis, the relative energy of the 3Zn^{II} configuration is set to 0.

respectively) were identical to that of $3\text{Zn}^{\text{II}}\text{-B}_2$ (i.e., Conformer-2) featuring three interfacial metal coordination sites (Figure S4f). Anomalous diffraction data collected at Cu^{II} (9.3 keV) and Zn^{II} (9.7 keV) K-edges confirmed that all interfacial ions were Zn^{II} (Figure S5a,b), corroborating our findings from solution analyses.

To probe whether the crystallographically observed structures also formed in solution, we performed solution small-angle X-ray scattering (SAXS) experiments. The samples contained 200 μM B_2 and 150 mM NaCl in a MOPS (pH 7.4) buffer solution in the presence of Cu^{II} or Zn^{II} only (600 μM each), as well as under competitive $\text{Cu}^{\text{II}}//\text{Zn}^{\text{II}}$ or $\text{Zn}^{\text{II}}//\text{Cu}^{\text{II}}$ (600 μM each) conditions (Figure S5c,d). In all cases, the scattering patterns showed linearities in Guinier analysis, consistent with B_2 dimers with a radius of gyration (R_g) of ~ 20.5 Å (Figure S5). We then compared the scattering patterns of Cu^{II} or Zn^{II} -containing samples with the theoretical scattering pattern derived from the $3\text{Zn}^{\text{II}}\text{-B}_2$ crystal structure (Figure S5c). There was a close agreement between the experimental SAXS pattern of Zn^{II} -bound B_2 ($\chi^2 \sim 0.8$) with the theoretical pattern, indicating that the $\text{Zn}\text{-B}_2$ complex formed in solution corresponds to the crystal structure of $3\text{Zn}^{\text{II}}\text{-B}_2$ (Conformer-2). In contrast, there was a large deviation in the case of the Cu^{II} -bound B_2 ($\chi^2 \sim 13.4$) from the simulated SAXS pattern of the $3\text{Zn}^{\text{II}}\text{-B}_2$ structure (Figure S5c). In fact, the pattern for Cu^{II} -bound B_2 more closely approximated that derived from the $1\text{Ni}^{\text{II}}\text{-B}_2$ crystal structure ($\chi^2 \sim 2.0$) (Figure S6), implying that the Cu^{II} -bound B_2 conformer may be structurally similar to the $1\text{Ni}^{\text{II}}\text{-B}_2$ complex (Conformer-3). Next, we compared the experimental SAXS patterns of B_2 subjected to competitive $\text{Cu}^{\text{II}}//\text{Zn}^{\text{II}}$ and $\text{Zn}^{\text{II}}//\text{Cu}^{\text{II}}$ conditions with the theoretical pattern for $3\text{Zn}^{\text{II}}\text{-B}_2$. Both samples showed very good agreement in the Kratky plots (Figure S5d), with corresponding χ^2 values of 1.4 and 2.2, respectively. These observations provide further evidence that B_2 selectively binds Zn^{II} in solution over Cu^{II} , adopting a solution conformation that is very similar to that observed crystallographically.

Tetrahedral Coordination Geometry as a Driving Force for the Zn^{II} Selectivity of B_2 . The trinuclear coordination motif in Conformer-2 is also available in solution

for Cu^{II} ions to bind yet the Zn^{II} -bound B_2 species completely outcompetes the Cu^{II} -bound form. This suggests that the coordination geometries observed in Conformer-2 must disfavor Cu^{II} binding. An analysis of the $3\text{Zn}^{\text{II}}\text{-B}_2$ structure indicates that the sidechains of the coordinating His and Glu residues cannot adopt alternative rotameric conformations due to steric occlusion by the proximal residues (Figure S7). This implies that the observed tetrahedral coordination geometries must be sufficiently rigid to prevent distortion into the preferred tetragonal/square-planar geometry by Cu^{II} coordination.

Since we could not experimentally evaluate the binding affinity of Cu^{II} ions to the coordination geometries observed in Conformer-2—because Cu^{II} binding likely induces the Conformer-3 geometry—we computationally evaluated the energetic outcome of substituting one of the Zn^{II} ions with Cu^{II} using density functional theory (DFT) calculations (Figure 6a–c). In these calculations, we modeled metal exchange reactions between $3\text{Zn}^{\text{II}}\text{-B}_2$ and two different $(2\text{Zn}^{\text{II}}+1\text{Cu}^{\text{II}})\text{-B}_2$ states (Model 1 and Model 2), wherein hydrated Cu^{II} replaces either a peripheral or the central Zn^{II} ion in the trinuclear site (Figure 6d,e). In these calculations, the positions of the coordinating ligands were fixed in light of their limited ability to explore alternative rotameric states (Figure S7). We found that the replacement of a Zn^{II} ion with Cu^{II} raised the overall energy of the systems in both Models 1 and 2, with particularly strong effects observed for the coordinatively saturated peripheral site in Model 1.

To further investigate the role of the primary coordination sphere(s) in Conformer-2 in imposing Zn^{II} selectivity over Cu^{II} , we investigated the metal-binding properties of B_2 variants in which each of the metal-coordinating His ligands were mutated to Ala (H60A, H63A, and H104A). Competitive metal binding assays showed that none of the B_2 variants could compete with Fura-2 for Zn^{II} binding (Figure 7a), while they retained picomolar dissociation constants in the case of Cu^{II} (Figure S8a and Table S1). Similarly, ESI-MS spectra showed that the B_2 variants could no longer coordinate three Zn^{II} ions (Figure 7b) and selectively bind Zn^{II} over Cu^{II} (Figure S8b), establishing that the structural integrity (and the precise

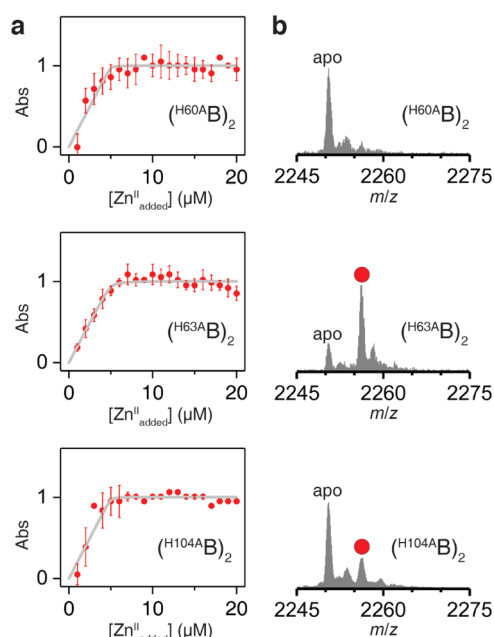


Figure 7. (a) Fura-2 competitive titration assay of H60A, H63A, and H104A mutants of B₂ with Zn^{II}. Changes in normalized Fura-2 absorbance at 335 nm (red data points) are plotted along with theoretical metal-binding isotherms (gray line) in the absence of the proteins. Due to the absence of competition by the Ala mutants, the absorbances of Fura-2 reach the saturation point at the concentration of Fura-2 (~5 μM). Experimental data points and error bars are presented as mean and standard deviation of three independent measurements. (b) ESI-MS spectra of Ala mutants of B₂ with Zn^{II}. Red circles represent the number of Zn^{II} ion bound to the mutants.

coordination geometry) of the trinuclear coordination mode in Conformer-2 is crucial for the observed Zn^{II} selectivity of B₂.

Zn^{II} Selectivity of B₂ in the *E. coli* Periplasmic Space.

We asked if the Zn^{II} selectivity of B₂ is also operative in the highly heterogeneous chemical environment of the cellular milieu. Specifically, we targeted the periplasmic space of *Escherichia coli* (*E. coli*) cells, which is more permissive in terms of metal/chemical exchange with the external medium compared to the cytosol and thus has proven to be particularly suitable for the development of artificial metalloproteins/metalloenzymes with *in vivo* functions.^{57,58,78–86} Conveniently, cytochrome *cb*₅₆₂ (parent of B₂) is equipped with an N-terminal signal sequence for Sec-dependent translocation into the periplasmic space,⁸⁷ whose oxidative environment enables disulfide bond formation for B₂ dimer assembly. In our experiments, B₂ was expressed in *E. coli* BL21 (DE3) cells, whereby the lysogeny broth (LB) growth medium was supplemented with 50 μM Cu^{II}, Zn^{II}, or a combination of Cu^{II} and Zn^{II} (Figure 8a). After 16 h of expression at 37 °C, B₂ components were extracted using osmotic cold shock and enriched through sequential filtration steps.^{56,58}

To determine the periplasmic assembly state of B₂, we separated proteinaceous components of the periplasmic extracts using SDS-PAGE and stained all proteins with Coomassie blue (Figure 8b) or *o*-dianisidine (Figure 8c)—a chromogenic peroxidase substrate which leads to selective staining of all heme proteins.⁸⁸ Compared to the control group that does not have B₂-encoded plasmids, considerable amounts of both the B₂ dimer and monomer were expressed regardless of metal supplements (Figure 8b). Heme staining further

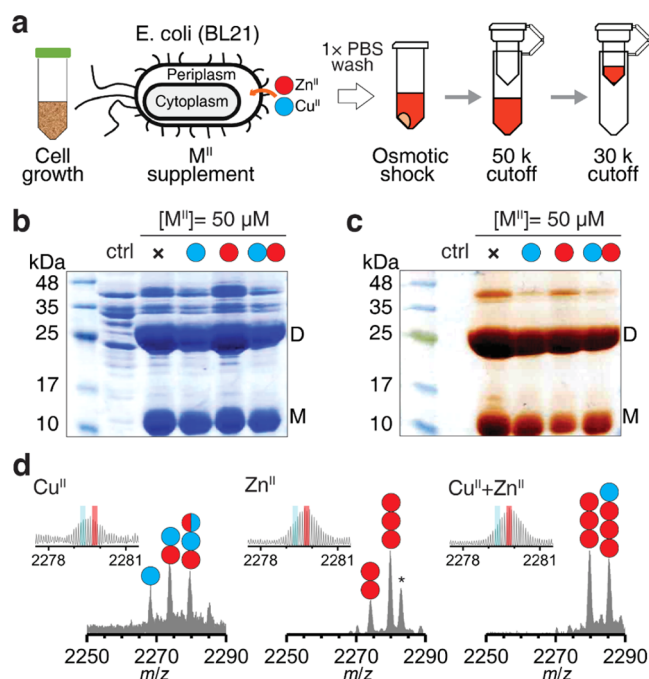


Figure 8. (a) Extraction protocol of M^{II}-B₂ complexes formed in the periplasm of *E. coli* BL21 cells. The cells expressing B₂ were grown in M^{II}-supplemented (50 μM) LB media, then underwent periplasmic extraction using 20% sucrose osmotic shock. Large aggregates were removed using 100:50 kDa centrifugal filters, and the M^{II}-B₂ complexes were enriched with a 30 kDa filter. (b, c) SDS-PAGE analysis of the periplasmic extracts containing M^{II}-B₂ complexes. Gels were stained with (b) Coomassie blue for proteinaceous contents and (c) *o*-dianisidine for heme proteins. (d) ESI-MS spectra of the periplasmic extracts from the cells grown with Cu^{II} (top), Zn^{II} (middle), and Cu^{II} + Zn^{II} (bottom). Filled circles represent the number of Cu^{II} (cyan) and Zn^{II} (red) ions bound to B₂ in the periplasm. Inlet ESI-MS spectra correspond to the *m/z* value of the 3M^{II}-B₂ species.

indicated that the dimer and monomer were properly folded through heme incorporation in the periplasm (Figure 8c). We analyzed the metalation state of B₂ in the heterogeneous cell extracts by ESI-MS. Interestingly, we detected 1Zn^{II}-, 2Zn^{II}-, and 3Zn^{II}-B₂ complexes even without metal supplementation (Figure S9a,b), a behavior previously not observed with the (AB)₂ variant.⁵⁶ Subsequent ICP-MS analyses indicated that the LB medium without metal supplements contains ~0.5 μM Cu^{II} and ~17 μM Zn^{II}. As a result, even in the Cu^{II}-enriched, non-Zn-supplemented medium, we observed that B₂ formed heterometallic Cu^{II} and Zn^{II} complexes (Figure 8d). Under both the Zn^{II}- and Zn^{II}/Cu^{II}-supplemented conditions, we detected 3Zn^{II}-B₂ as the major species (Figure 8d), confirming the Zn^{II} selectivity of B₂ in the complex chemical environment of the bacterial periplasm.

CONCLUSIONS

In the last decade, considerable progress has been made in the design and engineering of artificial metalloproteins with new or improved functions, ranging from enzymatic activities^{57,89–92} to unnatural reactions.^{78,79,84,93–96} Despite these advances, the rational design of protein structures to control metal binding and reactivity at different structural levels (primary, secondary and outer-coordination spheres) remains challenging. Against this backdrop, we have introduced a rational/probabilistic

design strategy, MASCOT, which is based on the construction of flexible yet covalently constrained protein-protein interfaces that are equipped with metal-binding motifs.⁴¹ While the flexibility of such interfaces allows for the thermodynamically controlled exploration of different metal coordination geometries, their conformationally constrained nature leads to structurally well-defined metal-bound states (*i.e.*, corresponding to distinct energy minima) which not only give rise to cooperative effects but are also only amenable to structural characterization. Using MASCOT, we had previously designed a disulfide-tethered dimer of cytochrome *cb*₅₆₂ termed (AB)₂, which was capable of selectively binding Co^{II} and Ni^{II} ions through the formation of coupled coordination sites that sterically excluded Cu^{II} binding.⁵⁶ Here, we demonstrated that (AB)₂ could be reengineered through a minimal amount of mutations to yield a second-generation variant, B₂, that displayed an altered selectivity for Zn^{II} binding over Cu^{II}, thus also overcoming the restrictions of the Irving–Williams series. This was achieved through the formation of an unusual tri-Zn coordination motif, which, to the best of our knowledge, does not have an analog among natural metalloproteins, although we note a recent study on the construction of a tetranuclear Zn^{II} cluster in a *de novo* designed helical bundle protein.⁹⁷ Despite the fact that this exact metal coordination motif was not foreseen, it has emerged as a result of the flexible design strategy. In light of the fact that MASCOT requires minimal genetic manipulation of a protein (for disulfide tethering or possibly genetic fusion) to create a flexible dimer, we believe that it will be particularly conducive to high-throughput/library design approaches for the discovery of new metal coordination motifs, metal-based reactivities, and selectivities.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.2c08050>.

Supporting methods; metal dissociation constants of B₂ variants (Tables S1); crystallization conditions (Tables S2); X-ray data collection and refinement statistics (Tables S3 and S4); theoretical and observed *m/z* values of B₂ complexes (Tables S5); experimental *m/z* values (Tables S6); competitive Zn^{II} titration between (AB)₂ and Fura-2 (Figure S1); Sedimentation velocity, charge-state distribution and crystal structure of apo-B₂ (Figure S2); cartoon representations of the 3Zn^{II}-B₂ crystal structure colored according to B-factors (Figure S3); Cu^{II} and Ni^{II} titration and ESI-MS spectra, and crystal structures (Figure S4); Guinier analysis (Figure S5); experimental SAXS profiles and normalized Kratky plots (Figure S6); possible rotamer orientations and average number of atomic contacts in 3Zn^{II}-B₂ (Figure S7); competitive Cu^{II} titration assays and ESI-MS spectra (Figure S8); and ESI-MS spectra of the periplasmic extract (Figure S9) (PDF)

■ AUTHOR INFORMATION

Corresponding Author

F. Akif Tezcan – Department of Chemistry and Biochemistry, University of California, San Diego, La Jolla, California 92093, United States; orcid.org/0000-0002-4733-6500; Email: tezcan@ucsd.edu

Author

Tae Su Choi – Department of Chemistry and Biochemistry, University of California, San Diego, La Jolla, California 92093, United States; orcid.org/0000-0002-1515-3275

Complete contact information is available at:

<https://pubs.acs.org/10.1021/jacs.2c08050>

Notes

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■ REFERENCES

- (1) Van Stappen, C.; Deng, Y.; Liu, Y.; Heidari, H.; Wang, J.-X.; Zhou, Y.; Ledray, A. P.; Lu, Y. Designing Artificial Metalloenzymes by Tuning of the Environment beyond the Primary Coordination Sphere. *Chem. Rev.* **2022**, *122*, 11974–12045.
- (2) Cvetkovic, A.; Menon, A. L.; Thorgersen, M. P.; Scott, J. W.; Poole, F. L.; Jenney Jr, F. E.; Lancaster, W. A.; Praissman, J. L.; Shanmukh, S.; Vaccaro, B. J.; Trauger, S. A.; Kalisiak, E.; Apon, J. V.; Siuzdak, G.; Yannone, S. M.; Tainer, J. A.; Adams, M. W. W. Microbial metalloproteomes are largely uncharacterized. *Nature* **2010**, *466*, 779–782.
- (3) Andreini, C.; Bertini, I.; Cavallaro, G.; Holliday, G. L.; Thornton, J. M. Metal ions in biological catalysis: from enzyme databases to general principles. *JBIC, J. Biol. Inorg. Chem.* **2008**, *13*, 1205–1218.
- (4) Song, W. J.; Sontz, P. A.; Ambroggio, X. I.; Tezcan, F. A. Metals in Protein–Protein Interfaces. *Annu. Rev. Biophys.* **2014**, *43*, 409–431.
- (5) Kluska, K.; Adamczyk, J.; Krężel, A. Metal binding properties, stability and reactivity of zinc fingers. *Coord. Chem. Rev.* **2018**, *367*, 18–64.
- (6) Ariöz, C.; Wittung-Stafshede, P. Folding of copper proteins: role of the metal? *Q. Rev. Biophys.* **2018**, *51*, No. e4.
- (7) Arnold, F. H.; Zhang, J.-H. Metal-mediated protein stabilization. *Trends Biotechnol.* **1994**, *12*, 189–192.
- (8) Shimizu, T.; Huang, D.; Yan, F.; Stranova, M.; Bartosova, M.; Fojtíková, V.; Martinková, M. Gaseous O₂, NO, and CO in Signal Transduction: Structure and Function Relationships of Heme-Based Gas Sensors and Heme-Redox Sensors. *Chem. Rev.* **2015**, *115*, 6491–6533.
- (9) Chang, C. J. Searching for harmony in transition-metal signaling. *Nat. Chem. Biol.* **2015**, *11*, 744–747.
- (10) Haas, K. L.; Franz, K. J. Application of Metal Coordination Chemistry To Explore and Manipulate Cell Biology. *Chem. Rev.* **2009**, *109*, 4921–4960.
- (11) Solomon, E. I.; Heppner, D. E.; Johnston, E. M.; Ginsbach, J. W.; Cirera, J.; Qayyum, M.; Kieber-Emmons, M. T.; Kjaergaard, C. H.; Hadt, R. G.; Tian, L. Copper Active Sites in Biology. *Chem. Rev.* **2014**, *114*, 3659–3853.
- (12) Shaanan, B. Structure of human oxyhaemoglobin at 2.1 Å resolution. *J. Mol. Biol.* **1983**, *171*, 31–59.

- (13) Holm, R. H.; Kennepohl, P.; Solomon, E. I. Structural and Functional Aspects of Metal Sites in Biology. *Chem. Rev.* **1996**, *96*, 2239–2314.
- (14) Rutledge, H. L.; Cook, B. D.; Nguyen, H. P. M.; Herzik, M. A.; Tezcan, F. A. Structures of the nitrogenase complex prepared under catalytic turnover conditions. *Science* **2022**, *377*, 865–869.
- (15) Ross, M. O.; MacMillan, F.; Wang, J.; Nisthal, A.; Lawton, T. J.; Olafson, B. D.; Mayo, S. L.; Rosenzweig, A. C.; Hoffman, B. M. Particulate methane monooxygenase contains only mononuclear copper centers. *Science* **2019**, *364*, 566–570.
- (16) Frandsen, K. E. H.; Simmons, T. J.; Dupree, P.; Poulsen, J.-C. N.; Hemsworth, G. R.; Ciano, L.; Johnston, E. M.; Tovborg, M.; Johansen, K. S.; von Freiesleben, P.; Marmuse, L.; Fort, S.; Cottaz, S.; Driguez, H.; Henrissat, B.; Lenfant, N.; Tuna, F.; Baldansuren, A.; Davies, G. J.; Lo Leggio, L.; Walton, P. H. The molecular basis of polysaccharide cleavage by lytic polysaccharide monooxygenases. *Nat. Chem. Biol.* **2016**, *12*, 298–303.
- (17) Huang, W.; Jia, J.; Cummings, J.; Nelson, M.; Schneider, G.; Lindqvist, Y. Crystal structure of nitrile hydratase reveals a novel iron centre in a novel fold. *Structure* **1997**, *5*, 691–699.
- (18) Kleefeld, O.; Frenkel, A.; Martin, J. M. L.; Sagi, I. Active site electronic structure and dynamics during metalloenzyme catalysis. *Nat. Struct. Biol.* **2003**, *10*, 98–103.
- (19) Waldron, K. J.; Robinson, N. J. How do bacterial cells ensure that metalloproteins get the correct metal? *Nat. Rev. Microbiol.* **2009**, *7*, 25–35.
- (20) Imlay, J. A. The Mismetallation of Enzymes during Oxidative Stress. *J. Biol. Chem.* **2014**, *289*, 28121–28128.
- (21) Smethurst, D. G. J.; Shcherbik, N. Interchangeable utilization of metals: New perspectives on the impacts of metal ions employed in ancient and extant biomolecules. *J. Biol. Chem.* **2021**, *297*, No. 101374.
- (22) Bloom, A. J. Metal regulation of metabolism. *Curr. Opin. Chem. Biol.* **2019**, *49*, 33–38.
- (23) Nianios, D.; Thierbach, S.; Steimer, L.; Lulchev, P.; Klostermeier, D.; Fetzner, S. Nickel quercetinase, a promiscuous metalloenzyme: metal incorporation and metal ligand substitution studies. *BMC Biochem.* **2015**, *16*, No. 10.
- (24) Bayliss, M. E.; Prescott, J. M. Modified activity of *Aeromonas* aminopeptidase: metal ion substitutions and role of substrates. *Biochemistry* **1986**, *25*, 8113–8117.
- (25) Gu, M.; Imlay, J. A. Superoxide poisons mononuclear iron enzymes by causing mismetallation. *Mol. Microbiol.* **2013**, *89*, 123–134.
- (26) Tottey, S.; Waldron, K. J.; Firbank, S. J.; Reale, B.; Bessant, C.; Sato, K.; Cheek, T. R.; Gray, J.; Banfield, M. J.; Dennison, C.; Robinson, N. J. Protein-folding location can regulate manganese-binding versus copper- or zinc-binding. *Nature* **2008**, *455*, 1138–1142.
- (27) Dudev, T.; Lim, C. Competition among Metal Ions for Protein Binding Sites: Determinants of Metal Ion Selectivity in Proteins. *Chem. Rev.* **2014**, *114*, 538–556.
- (28) Kiskeropoulos, E. C.; Griese, J. J.; Smith, Z. R.; Branca, R. M. M.; Schneider, C. R.; Högbom, M.; Shafaat, H. S. Key Structural Motifs Balance Metal Binding and Oxidative Reactivity in a Heterobimetallic Mn/Fe Protein. *J. Am. Chem. Soc.* **2020**, *142*, 5338–5354.
- (29) Foster, A. W.; Young, T. R.; Chivers, P. T.; Robinson, N. J. Protein metalation in biology. *Curr. Opin. Chem. Biol.* **2022**, *66*, No. 102095.
- (30) Carrell, C. J.; Wang, X.; Jones, L.; Jarrett, W. L.; Davidson, V. L.; Mathews, F. S. Crystallographic and NMR Investigation of Cobalt-Substituted Amicyanin. *Biochemistry* **2004**, *43*, 9381–9389.
- (31) Babor, M.; Greenblatt, H. M.; Edelman, M.; Sobolev, V. Flexibility of metal binding sites in proteins on a database scale. *Proteins: Struct., Funct., Bioinf.* **2005**, *59*, 221–230.
- (32) Edington, S. C.; Gonzalez, A.; Middendorf, T. R.; Halling, D. B.; Aldrich, R. W.; Baiz, C. R. Coordination to lanthanide ions distorts binding site conformation in calmodulin. *Proc. Natl. Acad. Sci. U.S.A.* **2018**, *115*, E3126–E3134.
- (33) Zaballa, M.-E.; Abriata, L. A.; Donaire, A.; Vila, A. J. Flexibility of the metal-binding region in apo-cupredoxins. *Proc. Natl. Acad. Sci. U.S.A.* **2012**, *109*, 9254–9259.
- (34) Eom, H.; Song, W. J. Emergence of metal selectivity and promiscuity in metalloenzymes. *J. Biol. Inorg. Chem.* **2019**, *24*, 517–531.
- (35) Frausto da Silva, J. J. R.; Williams, R. J. P. *The Biological Chemistry of the Elements*; Oxford University Press: Oxford, 2001.
- (36) Foster, A. W.; Osman, D.; Robinson, N. J. Metal Preferences and Metallation. *J. Biol. Chem.* **2014**, *289*, 28095–28103.
- (37) Williams, R. J. P. Metallo-Enzymes and Metallo-Proteins, Chemistry of. *Wiley Encycl. Chem. Biol.* **2007**, *0*, 1–8.
- (38) Irving, H.; Williams, R. J. P. Order of Stability of Metal Complexes. *Nature* **1948**, *162*, 746–747.
- (39) Gräve, K.; Griese, J. J.; Berggren, G.; Bennett, M. D.; Högbom, M. The *Bacillus anthracis* class Ib ribonucleotide reductase subunit NrdF intrinsically selects manganese over iron. *J. Biol. Inorg. Chem.* **2020**, *25*, 571–582.
- (40) Wang, S. C.; Dias, A. V.; Bloom, S. L.; Zamble, D. B. Selectivity of Metal Binding and Metal-Induced Stability of *Escherichia coli* NikR. *Biochemistry* **2004**, *43*, 10018–10028.
- (41) Rittle, J.; Field, M. J.; Green, M. T.; Tezcan, F. A. An efficient, step-economical strategy for the design of functional metalloproteins. *Nat. Chem.* **2019**, *11*, 434–441.
- (42) Zhu, J.; Avakyan, N.; Kakkis, A.; Hoffnagle, A. M.; Han, K.; Li, Y.; Zhang, Z.; Choi, T. S.; Na, Y.; Yu, C.-J.; Tezcan, F. A. Protein Assembly by Design. *Chem. Rev.* **2021**, *121*, 13701–13796.
- (43) Garzon-Rodriguez, W.; Yatsimirsky, A. K.; Glabe, C. G. Binding of Zn(II), Cu(II), and Fe(II) ions to Alzheimer's A β peptide studied by fluorescence. *Bioorg. Med. Chem. Lett.* **1999**, *9*, 2243–2248.
- (44) Hunt, J. A.; Ahmed, M.; Fierke, C. A. Metal Binding Specificity in Carbonic Anhydrase Is Influenced by Conserved Hydrophobic Core Residues. *Biochemistry* **1999**, *38*, 9054–9062.
- (45) Kakkis, A.; Gagnon, D.; Esselborn, J.; Britt, R. D.; Tezcan, F. A. Metal-Templated Design of Chemically Switchable Protein Assemblies with High-Affinity Coordination Sites. *Angw. Chem., Int. Ed.* **2020**, *59*, 21940–21944.
- (46) Ma, Z.; Jacobsen, F. E.; Giedroc, D. P. Coordination Chemistry of Bacterial Metal Transport and Sensing. *Chem. Rev.* **2009**, *109*, 4644–4681.
- (47) Maret, W. Metalloproteomics, metalloproteomes, and the annotation of metalloproteins. *Metallomics* **2010**, *2*, 117–125.
- (48) Rosenzweig, A. C. Copper Delivery by Metallochaperone Proteins. *Acc. Chem. Res.* **2001**, *34*, 119–128.
- (49) O'Halloran, T. V.; Culotta, V. C. Metallochaperones, an Intracellular Shuttle Service for Metal Ions. *J. Biol. Chem.* **2000**, *275*, 25057–25060.
- (50) Weiss, A.; Murdoch, C. C.; Edmonds, K. A.; Jordan, M. R.; Monteith, A. J.; Perera, Y. R.; Rodríguez Nassif, A. M.; Petoletti, A. M.; Beavers, W. N.; Munneke, M. J.; Drury, S. L.; Krystofiak, E. S.; Thalluri, K.; Wu, H.; Kruse, A. R. S.; DiMarchi, R. D.; Caprioli, R. M.; Spraggins, J. M.; Chazin, W. J.; Giedroc, D. P.; Skaar, E. P. Zn-regulated GTPase metalloprotein activator 1 modulates vertebrate zinc homeostasis. *Cell* **2022**, *185*, 2148–2163.
- (51) Jordan, M. R.; Wang, J.; Weiss, A.; Skaar, E. P.; Capdevila, D. A.; Giedroc, D. P. Mechanistic Insights into the Metal-Dependent Activation of Zn^{II}-Dependent Metallochaperones. *Inorg. Chem.* **2019**, *58*, 13661–13672.
- (52) Osman, D.; Foster, A. W.; Chen, J.; Svedaite, K.; Steed, J. W.; Lurie-Luke, E.; Huggins, T. G.; Robinson, N. J. Fine control of metal concentrations is necessary for cells to discern zinc from cobalt. *Nat. Commun.* **2017**, *8*, No. 1884.
- (53) Foster, A. W.; Pernil, R.; Patterson, C. J.; Scott, A. J. P.; Pålsson, L.-O.; Pal, R.; Cummins, I.; Chivers, P. T.; Pohl, E.; Robinson, N. J. A tight tunable range for Ni(II) sensing and buffering in cells. *Nat. Chem. Biol.* **2017**, *13*, 409–414.

- (54) Osman, D.; Martini, M. A.; Foster, A. W.; Chen, J.; Scott, A. J. P.; Morton, R. J.; Steed, J. W.; Lurie-Luke, E.; Huggins, T. G.; Lawrence, A. D.; Deery, E.; Warren, M. J.; Chivers, P. T.; Robinson, N. J. Bacterial sensors define intracellular free energies for correct enzyme metalation. *Nat. Chem. Biol.* **2019**, *15*, 241–249.
- (55) Young, T. R.; Martini, M. A.; Foster, A. W.; Glasfeld, A.; Osman, D.; Morton, R. J.; Deery, E.; Warren, M. J.; Robinson, N. J. Calculating metalation in cells reveals CobW acquires Co^{II} for vitamin B₁₂ biosynthesis while related proteins prefer Zn^{II}. *Nat. Commun.* **2021**, *12*, No. 1195.
- (56) Choi, T. S.; Tezcan, F. A. Overcoming universal restrictions on metal selectivity by protein design. *Nature* **2022**, *603*, 522–527.
- (57) Song, W. J.; Tezcan, F. A. A designed supramolecular protein assembly with in vivo enzymatic activity. *Science* **2014**, *346*, 1525–1528.
- (58) Medina-Morales, A.; Perez, A.; Brodin, J. D.; Tezcan, F. A. In Vitro and Cellular Self-Assembly of a Zn-Binding Protein Cryptand via Templated Disulfide Bonds. *J. Am. Chem. Soc.* **2013**, *135*, 12013–12022.
- (59) Deblonde, G. J. P.; Mattocks, J. A.; Park, D. M.; Reed, D. W.; Cotruvo, J. A.; Jiao, Y. Selective and Efficient Biomacromolecular Extraction of Rare-Earth Elements using Lanmodulin. *Inorg. Chem.* **2020**, *59*, 11855–11867.
- (60) Guffy, S. L.; Der, B. S.; Kuhlman, B. Probing the minimal determinants of zinc binding with computational protein design. *Protein Eng., Des. Sel.* **2016**, *29*, 327–338.
- (61) Akcapinar, G. B.; Sezerman, O. U. Computational approaches for de novo design and redesign of metal-binding sites on proteins. *Biosci. Rep.* **2017**, *37*, No. BSR20160179.
- (62) Zygiel, E. M.; Nolan, E. M. Transition Metal Sequestration by the Host-Defense Protein Calprotectin. *Ann. Rev. Biochem.* **2018**, *87*, 621–643.
- (63) Laitaoja, M.; Valjakka, J.; Jänis, J. Zinc Coordination Spheres in Protein Structures. *Inorg. Chem.* **2013**, *52*, 10983–10991.
- (64) Shumilina, E.; Dobrovol'ska, O.; Del Conte, R.; Holen, H. W.; Dikiy, A. Competitive cobalt for zinc substitution in mammalian methionine sulfoxide reductase B1 overexpressed in E. coli: structural and functional insight. *JBIC, J. Biol. Inorg. Chem.* **2014**, *19*, 85–95.
- (65) Andreini, C.; Banci, L.; Bertini, I.; Rosato, A. Zinc through the Three Domains of Life. *J. Proteome Res.* **2006**, *5*, 3173–3178.
- (66) Chen, A. Y.; Adamek, R. N.; Dick, B. L.; Credille, C. V.; Morrison, C. N.; Cohen, S. M. Targeting Metalloenzymes for Therapeutic Intervention. *Chem. Rev.* **2019**, *119*, 1323–1455.
- (67) Hanas, J. S.; Gunn, C. G. Inhibition of Transcription Factor IIIA-DNA Interactions by Xenobiotic Metal Ions. *Nucleic Acids Res.* **1996**, *24*, 924–930.
- (68) Huang, M.; Krepkiy, D.; Hu, W.; Petering, D. H. Zn-, Cd-, and Pb-transcription factor IIIA: properties, DNA binding, and comparison with TFIIIA-finger 3 metal complexes. *J. Inorg. Biochem.* **2004**, *98*, 775–785.
- (69) Bittel, D.; Dalton, T.; Samson, S. L. A.; Gedamu, L.; Andrews, G. K. The DNA Binding Activity of Metal Response Element-binding Transcription Factor-1 Is Activated in vivo and in Vitro by Zinc, but Not by Other Transition Metals. *J. Biol. Chem.* **1998**, *273*, 7127–7133.
- (70) Lee, J. H.; Lee, J. H.; Jung, S. H.; Hyun, T. K.; Feng, M.; Kim, J.-Y.; Lee, J.-H.; Lee, H.; Kim, J. S.; Kang, C.; Kwon, K.-Y.; Jung, J. H. Highly selective fluorescence imaging of zinc distribution in HeLa cells and Arabidopsis using a naphthalene-based fluorescent probe. *Chem. Commun.* **2015**, *51*, 7463–7465.
- (71) Komatsu, K.; Kikuchi, K.; Kojima, H.; Urano, Y.; Nagano, T. Selective Zinc Sensor Molecules with Various Affinities for Zn²⁺, Revealing Dynamics and Regional Distribution of Synaptically Released Zn²⁺ in Hippocampal Slices. *J. Am. Chem. Soc.* **2005**, *127*, 10197–10204.
- (72) Ghosh, P.; Maayan, G. A rationally designed peptoid for the selective chelation of Zn²⁺ over Cu²⁺. *Chem. Sci.* **2020**, *11*, 10127–10134.
- (73) Brodin, J. D.; Medina-Morales, A.; Ni, T.; Salgado, E. N.; Ambroggio, X. I.; Tezcan, F. A. Evolution of Metal Selectivity in Templated Protein Interfaces. *J. Am. Chem. Soc.* **2010**, *132*, 8610–8617.
- (74) Burgot, J.-L. *Ionic Equilibria in Analytical Chemistry*; Springer, 2012.
- (75) Weast, R. C. *CRC Handbook of Chemistry and Physics*; CRC Press, 1977.
- (76) Andreini, C.; Cavallaro, G.; Lorenzini, S.; Rosato, A. MetalPDB: a database of metal sites in biological macromolecular structures. *Nucleic Acids Res.* **2012**, *41*, D312–D319.
- (77) Putignano, V.; Rosato, A.; Banci, L.; Andreini, C. MetalPDB in 2018: a database of metal sites in biological macromolecular structures. *Nucleic Acids Res.* **2018**, *46*, D459–D464.
- (78) Jeschek, M.; Reuter, R.; Heinisch, T.; Trindler, C.; Klehr, J.; Panke, S.; Ward, T. R. Directed evolution of artificial metalloenzymes for in vivo metathesis. *Nature* **2016**, *537*, 661.
- (79) Liang, A. D.; Serrano-Plana, J.; Peterson, R. L.; Ward, T. R. Artificial Metalloenzymes Based on the Biotin–Streptavidin Technology: Enzymatic Cascades and Directed Evolution. *Acc. Chem. Res.* **2019**, *52*, 585–595.
- (80) Gao, L.; Zhang, Y.; Zhao, L.; Niu, W.; Tang, Y.; Gao, F.; Cai, P.; Yuan, Q.; Wang, X.; Jiang, H.; Gao, X. An artificial metalloenzyme for catalytic cancer-specific DNA cleavage and operando imaging. *Sci. Adv.* **2020**, *6*, No. eabb1421.
- (81) Naughton, K. J.; Treviño, R. E.; Moore, P. J.; Wertz, A. E.; Dickson, J. A.; Shafaat, H. S. In Vivo Assembly of a Genetically Encoded Artificial Metalloenzyme for Hydrogen Production. *ACS Synth. Biol.* **2021**, *10*, 2116–2120.
- (82) Grimm, A. R.; Sauer, D. F.; Polen, T.; Zhu, L.; Hayashi, T.; Okuda, J.; Schwaneberg, U. A Whole Cell E. coli Display Platform for Artificial Metalloenzymes: Poly(phenylacetylene) Production with a Rhodium–Nitrobindin Metalloprotein. *ACS Catal.* **2018**, *8*, 2611–2614.
- (83) Eda, S.; Nasibullin, I.; Vong, K.; Kudo, N.; Yoshida, M.; Kurbangalieva, A.; Tanaka, K. Biocompatibility and therapeutic potential of glycosylated albumin artificial metalloenzymes. *Nat. Catal.* **2019**, *2*, 780–792.
- (84) Ji, P.; Park, J.; Gu, Y.; Clark, D. S.; Hartwig, J. F. Abiotic reduction of ketones with silanes catalysed by carbonic anhydrase through an enzymatic zinc hydride. *Nat. Chem.* **2021**, *13*, 312–318.
- (85) Wittwer, M.; Markel, U.; Schiffls, J.; Okuda, J.; Sauer, D. F.; Schwaneberg, U. Engineering and emerging applications of artificial metalloenzymes with whole cells. *Nat. Catal.* **2021**, *4*, 814–827.
- (86) Bloomer, B. J.; Clark, D. S.; Hartwig, J. F. Progress, Challenges, and Opportunities with Artificial Metalloenzymes in Biosynthesis. *Biochemistry* **2022**, DOI: 10.1021/acs.biochem.1c00829.
- (87) Faraone-Mennella, J.; Tezcan, F. A.; Gray, H. B.; Winkler, J. R. Stability and Folding Kinetics of Structurally Characterized Cytochrome c-b562. *Biochemistry* **2006**, *45*, 10504–10511.
- (88) Claiborne, A.; Fridovich, I. Chemical and enzymic intermediates in the peroxidation of o-dianisidine by horseradish peroxidase. 1. Spectral properties of the products of dianisidine oxidation. *Biochemistry* **1979**, *18*, 2324–2329.
- (89) Song, W. J.; Yu, J.; Tezcan, F. A. Importance of Scaffold Flexibility/Rigidity in the Design and Directed Evolution of Artificial Metallo-β-lactamases. *J. Am. Chem. Soc.* **2017**, *139*, 16772–16779.
- (90) Yu, J.; Yang, J.; Seok, C.; Song, W. J. Symmetry-related residues as promising hotspots for the evolution of de novo oligomeric enzymes. *Chem. Sci.* **2021**, *12*, 5091–5101.
- (91) Khare, S. D.; Kipnis, Y.; Greisen, P., Jr.; Takeuchi, R.; Ashani, Y.; Goldsmith, M.; Song, Y.; Gallaher, J. L.; Silman, I.; Leader, H.; Sussman, J. L.; Stoddard, B. L.; Tawfik, D. S.; Baker, D. Computational redesign of a mononuclear zinc metalloenzyme for organophosphate hydrolysis. *Nat. Chem. Biol.* **2012**, *8*, 294–300.
- (92) Studer, S.; Hansen, D. A.; Pianowski, Z. L.; Mittl, P. R. E.; Debon, A.; Guffy, S. L.; Der, B. S.; Kuhlman, B.; Hilvert, D. Evolution of a highly active and enantiospecific metalloenzyme from short peptides. *Science* **2018**, *362*, 1285–1288.

(93) Huang, J.; Liu, Z.; Bloomer, B. J.; Clark, D. S.; Mukhopadhyay, A.; Keasling, J. D.; Hartwig, J. F. Unnatural biosynthesis by an engineered microorganism with heterologously expressed natural enzymes and an artificial metalloenzyme. *Nat. Chem.* **2021**, *13*, 1186–1191.

(94) Srivastava, P.; Yang, H.; Ellis-Guardiola, K.; Lewis, J. C. Engineering a dirhodium artificial metalloenzyme for selective olefin cyclopropanation. *Nat. Commun.* **2015**, *6*, No. 7789.

(95) Drienovská, I.; Rioz-Martínez, A.; Draksharapu, A.; Roelfes, G. Novel artificial metalloenzymes by in vivo incorporation of metal-binding unnatural amino acids. *Chem. Sci.* **2015**, *6*, 770–776.

(96) Miller, D. C.; Athavale, S. V.; Arnold, F. H. Combining chemistry and protein engineering for new-to-nature biocatalysis. *Nat. Synth.* **2022**, *1*, 18–23.

(97) Zhang, S.-Q.; Chino, M.; Liu, L.; Tang, Y.; Hu, X.; DeGrado, W. F.; Lombardi, A. De Novo Design of Tetranuclear Transition Metal Clusters Stabilized by Hydrogen-Bonded Networks in Helical Bundles. *J. Am. Chem. Soc.* **2018**, *140*, 1294–1304.

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